The Invariance Principle — problem 1.60

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Start with a point $S(a, b)$ of the plane with $0 < b < a$, and generate a sequence of points $S_n(x_n, y_n)$ according to the rule

$$x_0 = a, \quad y_0 = b, \quad x_{n+1} = \frac{2x_n y_n}{x_n + y_n}, \quad y_{n+1} = \frac{2x_{n+1} y_n}{x_{n+1} + y_n}.$$

Prove that there is a limiting point with $x = y$. Find this limit.